Enbridge Inc. [TSX:ENB]

Credit Risk Analysis as of December 31st, 2024

Credit Metrics Model Assumptions:

- Recovery rate = 50%
- Two Markov states
- Constant interest rates of the US Treasury yield in each year

	solvency	default
solvency	98.57%	1.43%
default	0.00%	100.00%

Figure 1.Transition matrix for one year

Maturity Date	US Treasury (%)	Enbridge Bond Yield in USD (%)	Spread (%)
2026-11-15	4.25%	4.92%	0.68%
2027-04-05	4.25%	4.80%	0.56%
2028-05-15	4.25%	4.95%	0.69%
2030-05-15	4.30%	5.12%	0.82%
2033-08-01	4.47%	5.42%	0.95%
2034-04-05	4.55%	5.32%	0.77%
2035-01-01	4.63%	5.17%	0.55%
	Average Spread:	0.72%	

Figure 3. Comparison of Enbridge Bond Yields to US Treasury Yields (2026–2035): Enbridge bonds yield 0.72% more on average than similar-maturity Treasuries. Note: as Enbridge bonds are priced in USD, the spread had to be computed relative to US Treasury yields.

Interest rate (% CAD T-Bonds)	3.14%	3.14%	3.14%	3.14%	3.14%	3.03%	3.10%	3.15%	3.20%	3.23%
Book value (Bn $ CAD)	218.97	218.97	218.97	218.97	218.97	218.97	218.97	218.97	218.97	218.97
Years to Maturity	1.00	2.00	3.00	4.00	5.00	6.00	7.00	8.00	9.00	10.00
Exercise price, K=L (Bn $ CAD)	154.45	158.72	163.21	168.13	173.55	179.41	185.68	192.27	199.13	206.19
Sigma s (%) (Five-year average)	29.66%	29.66%	29.66%	29.66%	29.66%	29.66%	29.66%	29.66%	29.66%	29.66%
Sigma a (%)	9.35%	9.38%	9.46%	9.61%	9.83%	10.11%	10.43%	10.79%	11.18%	11.59%
Probability of default=N(-d2)	0.00%	0.23%	1.11%	2.67%	4.82%	7.85%	10.66%	13.58%	16.52%	19.42%

Figure 2. shows the fifth iteration of the Merton model for every maturity

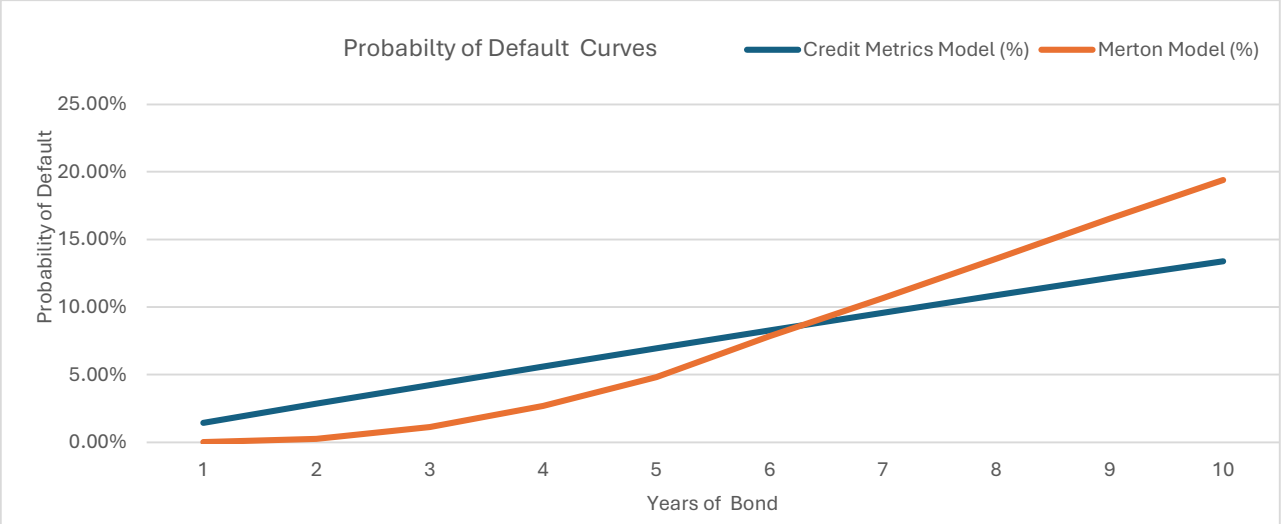

Figure 4. The probability of default over 10 years is estimated by the Credit Metrics Model and the Merton Model.

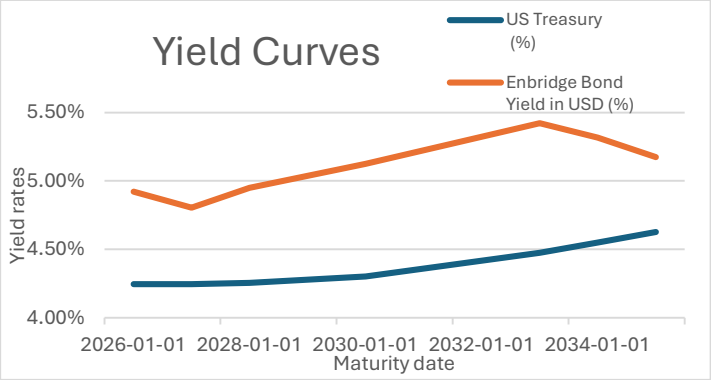

Figure 5. The chart compares the yield curves of Enbridge bonds and US Treasury securities for maturities from 2026 to 2035.

Data Sources: Bank of Canada, Business Insider website, Google Finance (embedded in Excel), St. Louis Fed website, 2024 Enbridge Annual Report